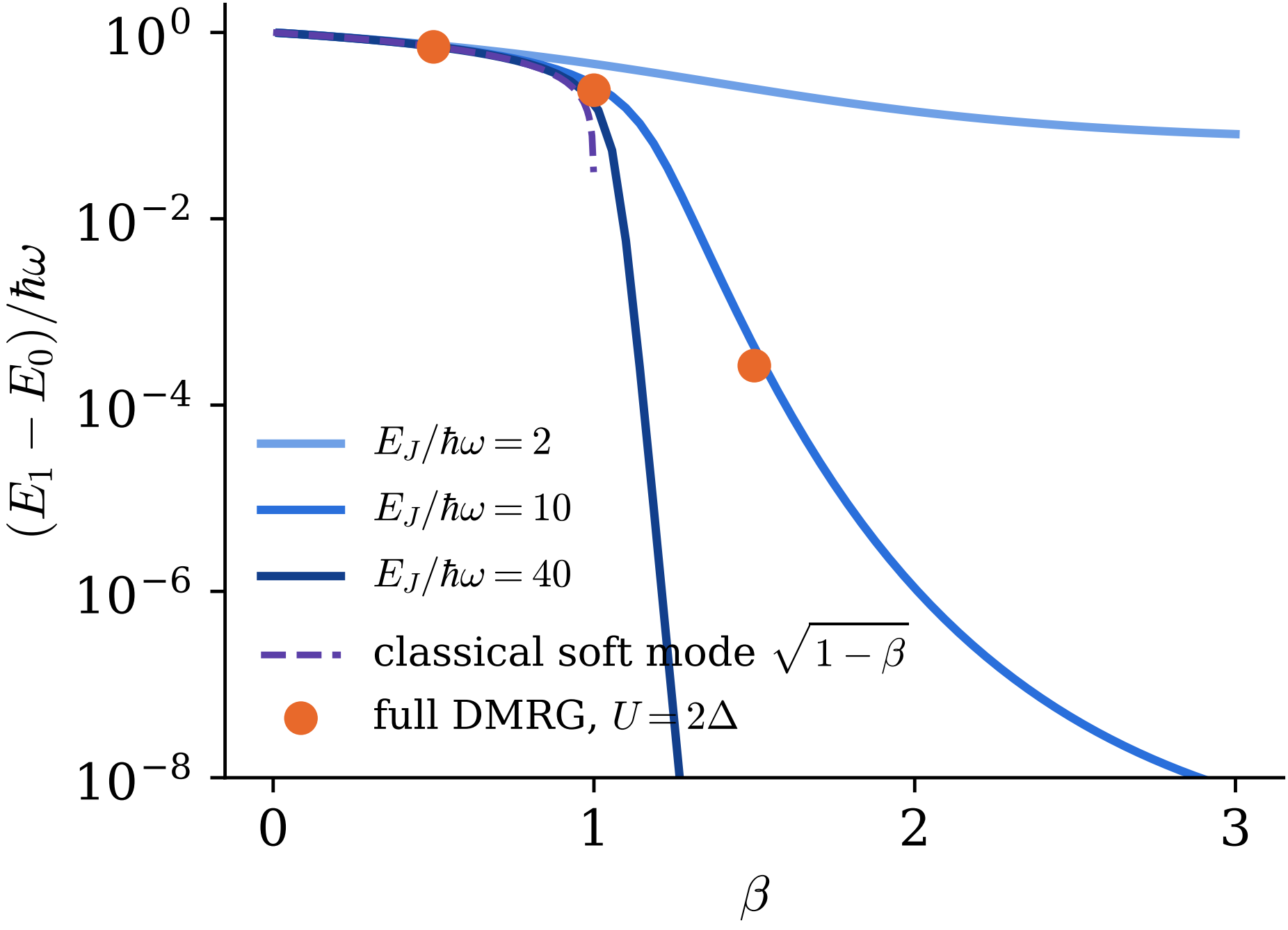
10^0
10^{-2}
10^{-4}
10^{-6}
10^{-8}
$(E_1 - E_0)/\hbar\omega$
0
1
2
3
β
$E_J/\hbar\omega = 2$
$E_J/\hbar\omega = 10$
$E_J/\hbar\omega = 40$
classical soft mode $\sqrt{1-\beta}$
full DMRG, $U = 2\Delta$